

Detailed changes

KiCad engine: 9.0.7. Complete readable register of changes.json. UUIDs identify objects in the PCB file.

Ground-zone changes

Zone UUID	Layer	Clearance before / after (mm)
a0eae8ca-36b8-4c27-901c-011db39c70fb	F.Cu	0.8 / 0.3
e51d2154-5fba-4386-a8ed-fa0b8de98b01	B.Cu	0.8 / 0.3

Pad connection

J301 pad 1 | Net: GND | Connection: solid

Eliminate isolated thermal spoke after ground refill; maintain continuous supply return.

Register totals

22 accepted trace-width changes; 40 added GND vias; 7 rejected trial width changes restored to original widths.

No components were moved or removed. Outline, cutouts and schematic connections were preserved. This register documents geometry changes; it does not establish measured thermal or electrical performance.

Accepted trace changes

Wider segments retained after clearance validation. Lengths are shown to six decimal places.

Track UUID	Net	Before / after width (mm)	Length (mm)
31ef5721-37c4-4afb-a183-91e3dbcd6aab	+5V_2	1.0 / 1.5	1.430000
8fe0f9db-3934-486b-b89d-d72e3d65f1bd	+5V_2	1.0 / 1.5	5.552900
1d886611-bd6e-4bd3-89fa-fe7d860c1dc4	+5V_2	1.0 / 1.5	4.875000
6a4d7f8d-76c3-4099-9ecd-1760d642f191	+5V_2	1.0 / 1.5	0.728320
c228a6d9-6566-492a-9b8f-61c2f1f1c938	+5V_2	1.0 / 1.5	1.428356
ddcf7a62-9421-440f-ab91-25352fdc545b	+5V_2	1.0 / 1.5	13.590000
38a982f1-31e3-4093-b7da-6f7d5a66890a	+5V_1	1.0 / 1.5	12.525000
7b42af86-8700-4ae5-87a2-bdae6c66ada5	+5V_1	1.0 / 1.5	7.276129
0c479267-2ed6-4ae2-a3ff-444148215c8c	+5V_1	1.0 / 1.5	1.682914
3e047d4c-6501-4dee-a039-87f970b2bc88	+5V_1	1.0 / 1.5	0.721249
79876056-91ae-4bd8-a9f7-9bf3e1b5d42d	+5V_1	1.0 / 1.5	1.343503
8b07e27b-5b8e-47e0-a421-9c9188633edd	+5V_1	1.0 / 1.5	0.862670
a100604c-2fe2-4a07-bb8b-b0f2b73a4795	+5V_1	1.0 / 1.5	6.127500
af8104fb-2127-4010-826c-19c26e32aef8	+5V_1	1.0 / 1.5	0.025000
b49ec555-dbeb-4bc7-8021-23aae4556453	+5V_1	1.0 / 1.5	7.110000
be5ffec7-982b-4e36-9b7a-fda47f46750c	+5V_1	1.0 / 1.5	36.420000
ea8b7dc0-9f6a-40dc-a41f-a371bb869255	+5V_1	1.0 / 1.5	0.042426
f1580c28-4462-418c-a6aa-7755514cde45	+5V_1	1.0 / 1.5	2.900000
32cc0a4c-58c2-4bfb-bd9e-dc7f79261db2	+UBEC_OUT	0.8 / 1.2	0.992500
c2030e9c-479f-4819-afa8-48d06526600c	+UBEC_OUT	0.8 / 1.2	2.973384
59d9cac1-4bbb-4a7c-98b0-3134c5ca3544	+UBEC_OUT	0.8 / 1.2	3.719382
86840676-2275-4376-aa27-94664568face	+UBEC_OUT	0.8 / 1.2	1.535000

Added ground vias

All 40 vias connect GND between F.Cu and B.Cu. Coordinates and dimensions are millimetres.

Via UUID	X	Y	Diameter	Drill
4f058209-c8e0-4b9f-b7aa-72ac596e9f02	190	104	0.8	0.4
5103b338-409e-489a-b3b5-4ae4d1853f16	190	110	0.8	0.4
a6444da1-77a7-4621-9709-567f652d4c49	190	116	0.8	0.4
252e0d64-be67-4016-a4a2-41b4f9598200	190	122	0.8	0.4
d03ba79b-2f0c-4446-963f-7663e0cc5c0d	190	134	0.8	0.4
b4dd7a33-aef1-4a96-b11f-eeef50c902d8	190	140	0.8	0.4
f5109881-d996-41fa-a389-535b86e29b68	190	146	0.8	0.4
1c30795d-1eeb-4a58-a370-36d789f87e49	190	164	0.8	0.4
d711969d-a0b2-45cb-bb32-4528ecd0d391	190	170	0.8	0.4
da337b94-2b8e-43b0-8fa6-fed4939105ff	196	104	0.8	0.4
a62c1e2c-7fc4-4343-b256-057f18a1f4e2	196	110	0.8	0.4
7e1838f3-ef65-4b3e-8d48-edabee23e854	196	116	0.8	0.4
ed7dad48-4d0a-41ae-8c1c-96bab078ccab	196	122	0.8	0.4
03dadaa3-797c-4d17-ba16-51785dd43cf3	196	134	0.8	0.4
bf96cb1e-32a4-4a60-a619-a5ad34a8ed71	196	140	0.8	0.4
7a7d458e-08df-4d22-8090-dac7046168c6	196	146	0.8	0.4
ab091ba1-f8e1-4725-9f13-990401a697a5	196	158	0.8	0.4
0c003798-fe19-4071-bedf-bb88ac67e76c	196	164	0.8	0.4
6c04c590-fe63-46f1-a685-370f970ca067	202	110	0.8	0.4
05e15dcb-8308-4fa4-a1a2-068f5a53d24d	202	134	0.8	0.4
02cf8eac-71b3-4944-8bb9-82d6eb872edf	202	140	0.8	0.4
83f1908e-48be-4d0c-8dcb-c7663d5c295d	202	146	0.8	0.4
a4af87c3-5e21-49e6-a645-8c971389039f	202	158	0.8	0.4
d937447c-dd04-42aa-90ed-05df7d6025d4	208	158	0.8	0.4
beb62826-1421-4f06-a1d8-d84699df7e99	208	170	0.8	0.4
27d15ff4-b6c6-47e1-a1b9-44a07a532ed2	214	140	0.8	0.4
afaafc92-153c-4351-af02-d25854cb5626	214	170	0.8	0.4
852bd6c5-30c3-4aae-9d8b-77855ad1a145	220	128	0.8	0.4
8d4ed9b1-c5d5-4d54-adae-d1c710ec256d	226	116	0.8	0.4
747cefeb-d623-4cd5-b31c-3fa29de4c4db	232	116	0.8	0.4
e33ae8e0-2945-475b-8e82-2f20a26c2bb5	232	128	0.8	0.4
25e6cc08-bdda-4d24-be4a-bb8f0fb3299d	232	170	0.8	0.4
b35144c2-46f7-4b66-b311-5689b9287cce	238	146	0.8	0.4

Via UUID	X	Y	Diameter	Drill
8337b717-4951-42d9-9adb-6c621894dd49	238	170	0.8	0.4
eba2fa26-b37a-4e66-8c9e-c4def7609f1d	244	122	0.8	0.4
23c66fe4-8d05-4c6a-a3c4-d33e21650c0b	244	152	0.8	0.4
38231787-8827-4f4f-aed7-17a2558a775c	244	158	0.8	0.4
cbd844e1-5888-4da9-8527-a3061e860717	250	164	0.8	0.4
d236f57e-b7d1-4b3a-99c8-a516b9ed016c	280	170	0.8	0.4
69ed1ec7-236f-429c-b66a-0b829ff3a820	292	158	0.8	0.4

Rejected trial changes

These proposed width increases did not pass clearance checks and were reverted. The after values below are rejected trial values, not the final board widths.

Track UUID	Net	Before / after width (mm)	Length (mm)
33dcc04a-68da-46ac-abc9-14f14b844d02	+5V_2	1.0 / 1.5	3.068000
d59b67b2-0072-4cc7-8c16-a5a8bba0c551	+5V_2	1.0 / 1.5	5.400000
f29ac1e3-2c80-4949-89cc-0f2107b0e8a6	+5V_1	1.0 / 1.5	1.819800
109b701d-eebe-41b2-a856-4d2e9f344b4c	+5V_1	1.0 / 1.5	6.140000
317e9d76-75eb-488f-90a4-51a9b8c9a45b	+5V_1	1.0 / 1.5	8.686807
432518a4-6154-4fb2-a027-37da3119bf04	+5V_1	1.0 / 1.5	8.950000
42cd3acf-8253-40e7-8f70-5f27f708a503	+UBEC_OUT	0.8 / 1.2	21.855000